Student Number

PEM

2020 Trial Higher School Certificate Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning
- and/or calculations

Total marks: 100

Section I – 10 marks (pages 2-6)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 9-34)

- Attempt Questions 11-34
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple -choice answer sheet for Questions 1-10

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1 The function $f(x)=3-2x$ has domain $[-1,2)$. The range of f is

A $[-1,2)$

B $[1,7)$

C $(-1,2]$

D $(-1,5]$

2 Given $f(x)=\log_e(x-3)$ and $g(x)=x+4$, the domain of $f(g(x))$ is

A $x\geq -4$

B $x>-1$

C $x>3$

D $x\geq 3$

3 It is given that $f'(x)=2g'(x)-1$, $f(0)=1$ and $g(0)=2$. Which of the following is true?

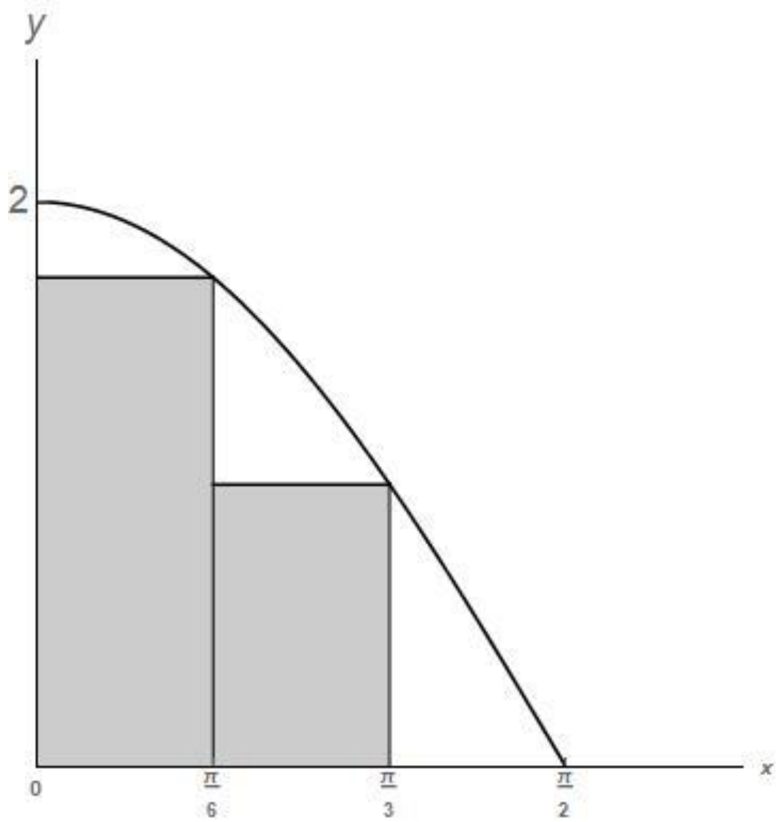
A $f(x)=1$

B $f(x)=2g(x)-x$

C $f(x)=2g(x)-x-3$

D $f(x)=\left[g(x)\right]^2-x-3$

4 The area under the curve $y = 2\cos x$, shown below, is approximated by two rectangles.



The value of the approximation is

A 1

B $\dfrac{\pi\left(\sqrt{3}+1\right)}{6}$

C $\left(\sqrt{3}+1\right)$

D $2\left(\dfrac{\pi}{6}+\dfrac{\pi}{3}\right)$

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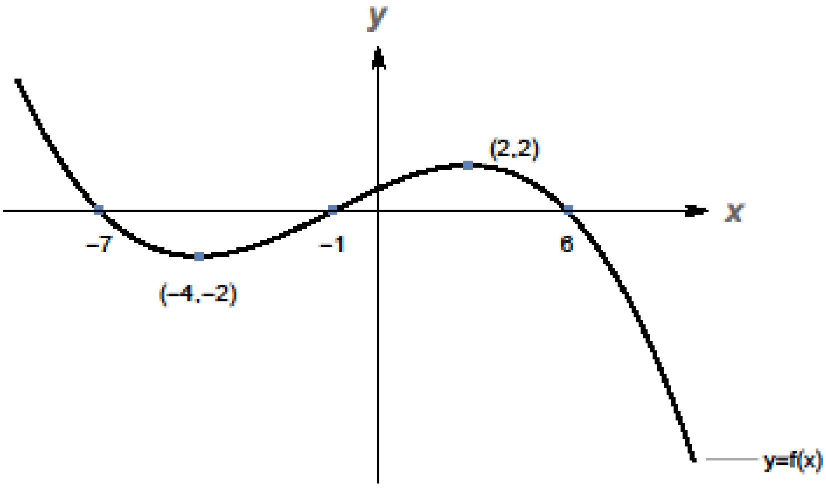

For the graph $y = f(x)$ shown above, $f'(x)$ is negative when

A $-7 < x < -1$ or $x > 6$

B $x < -4$ or $x > 2$

C $-4 < x < 2$

D $-2 < x < 2$

6 The discrete random variable X has the following probability distribution.

X	0	1	2
$P(X = x)$	a	b	0.3

Given that $E(X) = 0.8$, then

A $a = 0.5$ and $b = 0.2$

B $a = 0.2$ and $b = 0.5$

C $a = 0.3$ and $b = 0.4$

D $a = 0.3$ and $b = 0.2$

7 Let $f: R \to R$ be a differentiable function such that

- $f'(-2) = 0$; and
- $f'(x) > 0$ if $x \neq -2$.

At $x = -2$, the graph of $y = f(x)$ has a

A minimum turning point;

B maximum turning point;

C vertical asymptote;

D horizontal point of inflection.

8 The random variable X is distributed normally with $\mu = 12$ and $\sigma = 2$, and the random variable Z has the standard normal distribution.

Which of the following is true?

A $P(X < 9) = P(Z > 1.5)$

B $P(X < 9) = P(0 < Z < 1.5)$

C $P(X < 9) = P(-1.5 < Z < 1.5)$

D $P(X < 9) = P(-1.5 < Z < 0)$

9 At time t a particle has displacement and acceleration functions $x(t)$ and $a(t)$.

For which of the following functions is $x(t) \equiv a(t)$?

A $x(t) = 3\sin(t) - e^{t}$

B $x(t) = 3\cos(t) - e^{-t}$

C $x(t) = e^{t} - e^{-t}$

D $x(t) = 3\sin(t) - 3\cos(t)$

10 The sample space when a fair die is rolled is $\{1,2,3,4,5,6\}$. Each outcome is equally likely.

For which pair of events, M and N, are M and N independent?

A $A = \{2,3\}$ and $B = \{2,3,5,6\}$

B $A = \{1,3,5\}$ and $B = \{2,4,6\}$

C $A = \{1,2,3\}$ and $B = \{3,4,5\}$

D $A = \{1,3,5\}$ and $B = \{3,6\}$

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Trial Higher School Certificate Examination

Mathematics Advanced

Section II Answer Book

90 marks
Attempt Questions 11-34
Allow about 2 hours and 45 minutes for this section

Instructions

Answer the questions in the space provided. These spaces provide guidance for the expected length of response.

Your response should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of this booklet.

If you use this space, clearly indicate which question you are answering.

Please turn over

Question 11 (2 marks)

If $f(x) = \log_e(x^2 + 1)$, calculate $f'(-1)$. **2**

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Question 12 (3 marks)

Differentiate $e^{x \sin x}$. **3**

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Question 13 (2 marks)

At the point $(2,-4)$ on the curve $y=f(x)$, the tangent has equation $y=2-3x$. **2**

Determine the equation of the tangent to the curve $y=f(x+1)-2$ at the point $(1,-6)$

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Question 14 (2 marks)

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Find an anti-derivative of $\dfrac{2}{2-3x}$.

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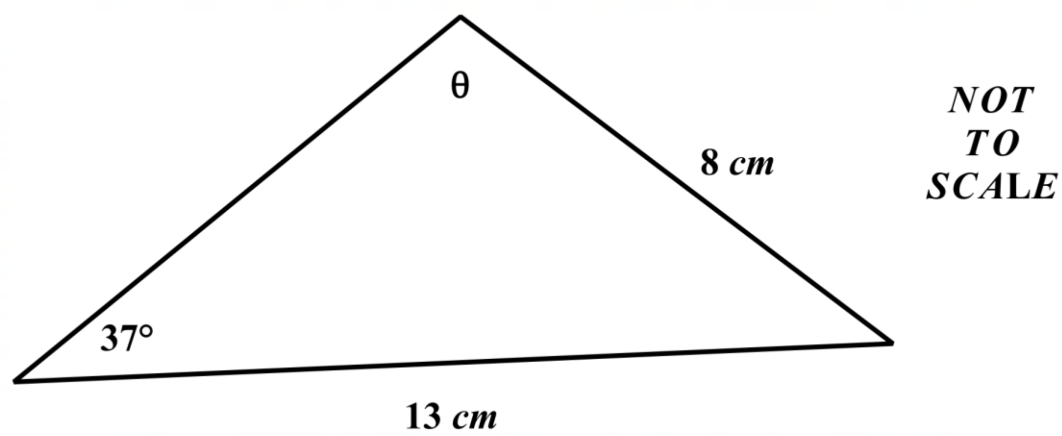
Question 15 (3 marks)

The diagram shows an acute angled triangle with two known side lengths, $8\,cm$ and $13\,cm$, and one known angle of 37°. **3**



Determined the value of θ, correct to the nearest minute.

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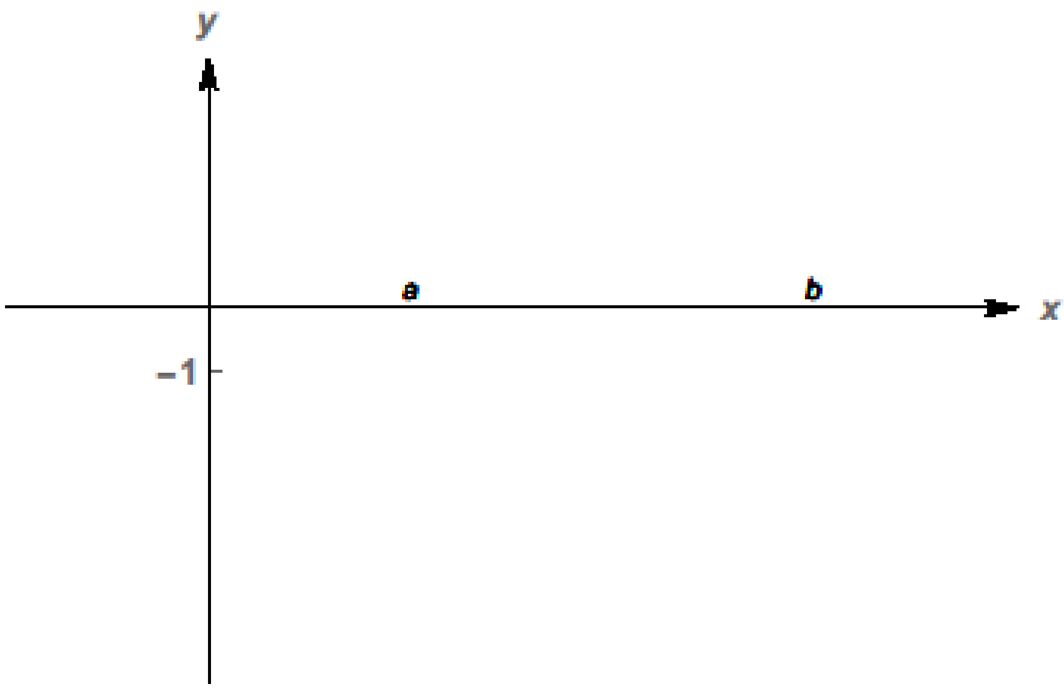
Question 16 (3 marks)

Over the domain $[a,b]$, function f satisfies the following conditions:

$$f(x) < -1,\ f'(x) < 0 \text{ and } f''(x) > 0.$$

(a) Sketch a possible graph of $y = f(x)$ for the domain $[a,b]$. **2**



(b) State the minimum value of the function $g(x) = |1 + f(x)|$ over the domain $[a,b]$. **1**

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Question 17 (4 marks)

A bag contains 6 identical balls, numbered 1, 2, 3, 4, 5 and 6. A ball is randomly selected, then, without replacement, a second ball is randomly selected.

(a) What is the probability that the first ball selected is numbered 3 and the second ball selected is numbered 4? **1**

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(b) The numbers on the two selected balls are added together. What is the probability that the sum of the numbers on the two balls is 7? **1**

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(c) If it is given that the sum of the numbers on the two balls is 7, determine the probability that the second ball drawn is numbered 4. **2**

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Question 18 (5 marks)

(a) Differentiate $(\sin x)^2$. **2**

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(b) Hence calculate $\int_0^{\pi/4} (\sin x + \cos x)^2 \, dx$, leave your answer in exact form. **3**

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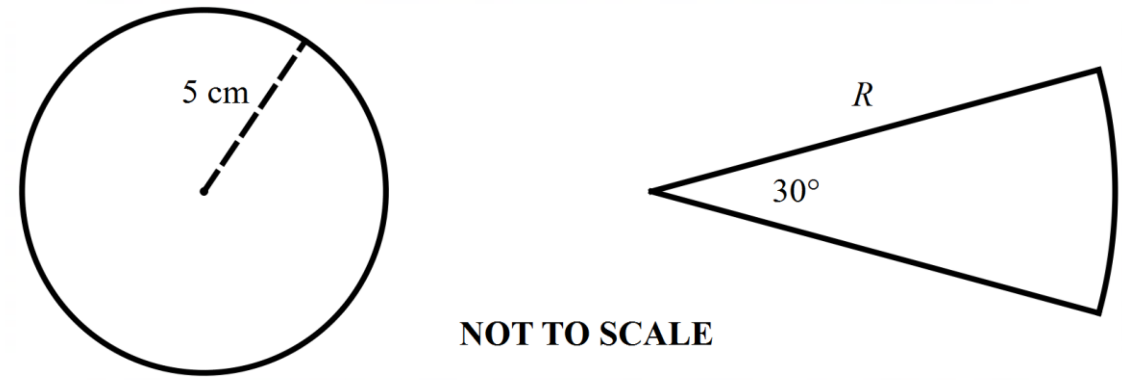
Question 19 (2 marks)

A circle of radius 5 cm and a sector, of radius $R\,cm$ has a central angle of 30°, as shown. **2**



If the circle and the sector have equal areas, determine the radius, R, of the sector.

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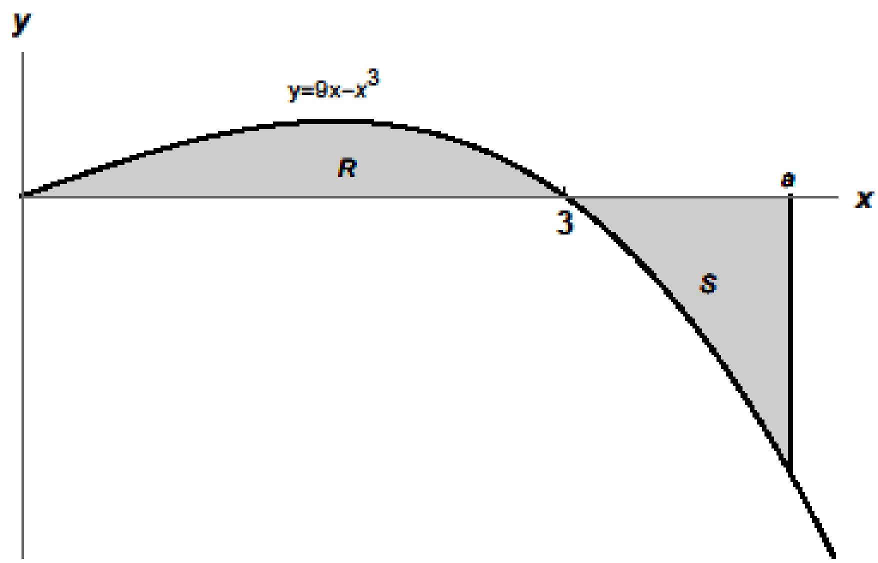
Question 20 (3 marks) **3**



The above diagram shows the cubic curve $y = 9x - x^3$, with domain $[0, 4.5]$.

The curve crosses the $x-$axis at $x = 0$ and $x = 3$.

The diagram shows two bounded regions:

R bounded by $y = f(x), y = 0, x = 0$ and $x = 3$

S bounded by $y = f(x), y = 0, x = 3$ and $x = a$

It is given that the $a > 3$. Determine the value of a such that regions R and S have the same area.

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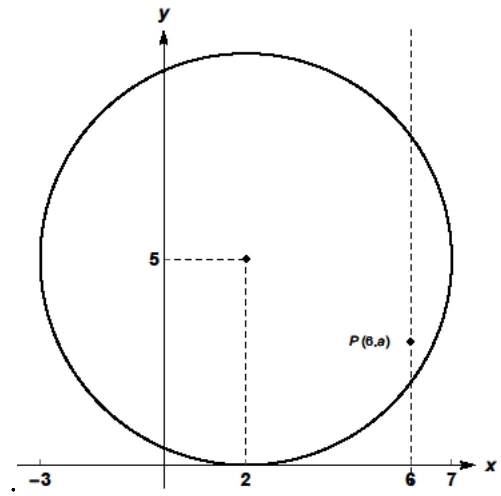
Question 21 (4 marks)

(a) Determine the equation of the circle shown below: **2**



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(b) The diagram above shows the point $P(6, a)$ which lies **inside** the circle. Determine the possible values of a **2**

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Question 22 (3 marks)

A drone is used during the filming of a television show. The drone leaves its base station, at point O, and flies $50\,m$ on a bearing of $045^{\circ}T$ to point A. It then changes direction to $130^{\circ}T$ and flies a further $90\,m$ to point B. **3**

To the nearest metre, calculate how far the drone is from base when it reaches point B.

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Question 23 (3 marks)

The function $f\left(x\right)=A\sin x+B$, where A and B are constants, passes through the points $P\left(\frac{\pi}{6},-3\right)$ and $Q\left(\frac{\pi}{2},1\right)$. **3**

By solving a pair of simultaneous equations, or otherwise, determine the range of f.

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Question 24 (5 marks)

(a) Prove that, for $\sin x\neq\pm1$: **2**

$$\frac{1}{1-\sin x}+\frac{1}{1+\sin x}\equiv 2\sec^2 x.$$

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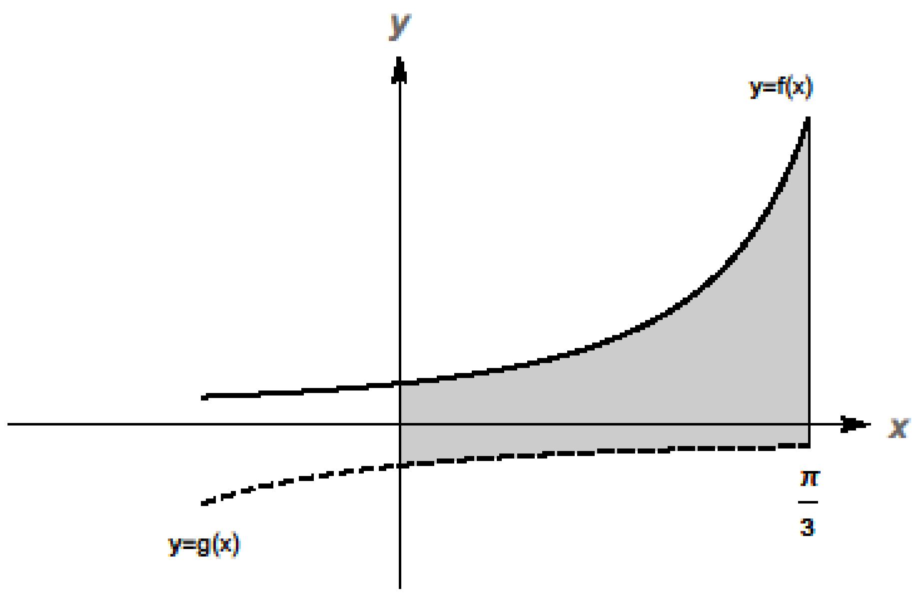
Question 24 continues on the next page

Question 24 (continued)

(b) **3**



The region shown above represents the area between the curves $f\left(x\right)=\dfrac{1}{1-\sin x}$ and $g\left(x\right)=\dfrac{-1}{1+\sin x}$, for the domain $\left[0,\dfrac{\pi}{3}\right]$. Calculate the area between the two curves.

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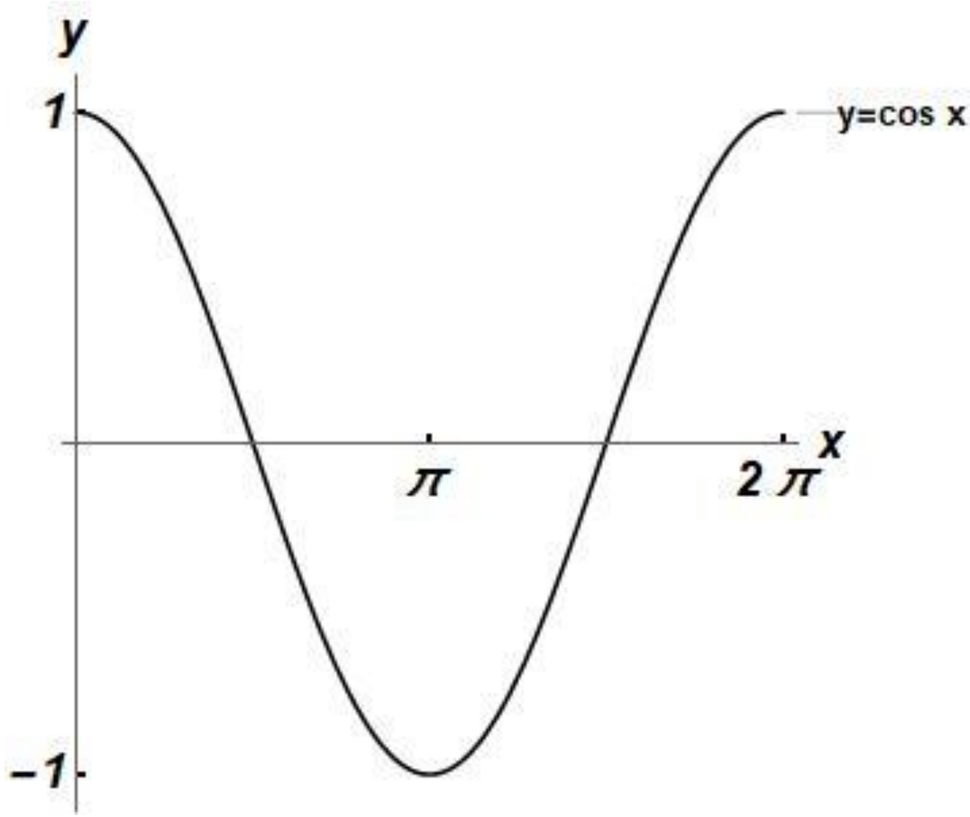
Question 25 (4 marks)

(a) On the set of axes below, draw the graph of $y = e^{-x}$. **1**



(b) State the number of solutions of the equation $\cos x = e^{-x}$ over the domain $[0, 2\pi]$. **1**

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(c) Determine the number of solutions of the equation $e^{x}\cos(2x) + 1 = 0$ over the domain $[0, 2\pi]$. **2**

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Question 26 (5 marks)

Let X be a discrete random variable with probability density function given by

$$P(X = x) = \frac{1}{2^x} \qquad \text{for} \quad x = 1, 2, 3, \ldots$$

(a) Show that the cumulative distribution function is given by **2**

$$F_X(x) = 1 - 2^{-x}.$$

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(b) Find $P(2 < X \leq 5)$ **2**

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(c) Find $P(X > 4)$ **1**

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Question 27 (6 marks)

(a) Determine the stationary points of $f(x) = -x^3 + 9x^2 - 24x + 16$ and their nature. **3**

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Question 27 continues on the following page

Question 27 (Continued)

(b) Labelling the stationary points and intercepts with the coordinate axes, sketch the graph of $y = f(x)$ over the domain $[0, 6]$. **2**

(c) State the global minimum of $y = f(x)$ over the domain. $[0, 6]$ **1**

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Question 28 (6 marks)

The monthly sales, X (in units of $\$100,000$), of a business are modelled by the two probability density functions:

$$g(x)=\begin{cases}2x & \text{for } 0\le x\le 1\\ 0 & \text{otherwise}\end{cases} \qquad \text{OR} \qquad f(x)=\begin{cases}kx^3\left(1-x^2\right) & \text{for } 0\le x\le 1\\ 0 & \text{otherwise}\end{cases}$$

(a) Determine the value of k in the second model. **2**

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(b) Show that the two models have the same median sales value. **4**

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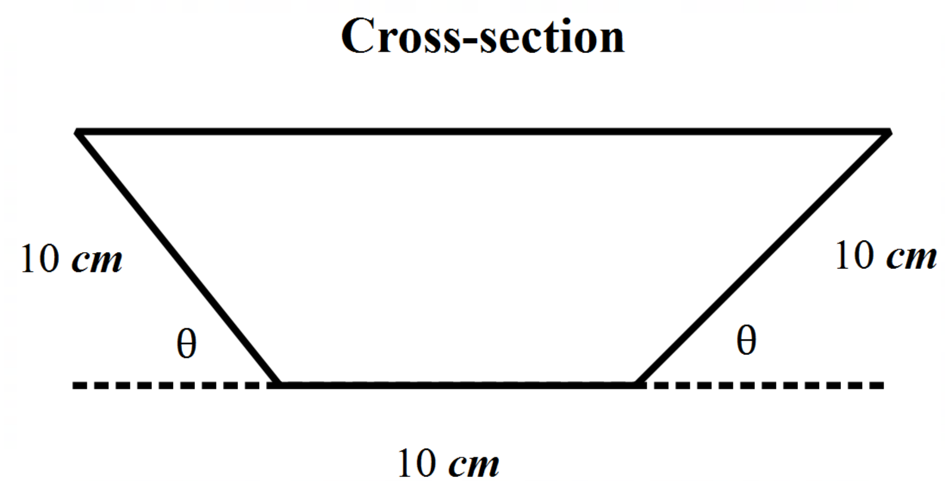
Question 29 (6 marks)

(a) The diagram below shows a 30 cm wide rectangular piece of metal, which is bent to form a gutter for the flow of water. The cross-section of a gutter is a trapezium. **2**



Show that the cross-sectional area $A(\theta)$, as a function of θ, is given by:

$$A(\theta) = 100\sin\theta(1+\cos\theta)$$

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Question 29 continues on the following page

Question 29 (Continued)

(b) Find the angle θ that maximizes the cross-sectional area. **4**

Question 30 (2 marks)

Sarita opens an annuity account, which requires contributions to be made at the end of each period. The table below shows the future of an annuity of $1 for various interest rates per period. **2**

Future value of an annuity of $1

Periods	**Interest rate per period**			
	0.25%	***0.75%***	***1.5%***	***3%***
2	2.0025	2.0075	2.015	2.03
6	6.0376	6.1136	6.2296	6.4684
12	12.1664	12.5076	13.0412	14.1920
24	24.7028	26.1885	28.6335	34.4265

To save for a computer for university, Sarita makes regular deposits at the end of each month into an annuity account that pays 3% p.a. . The annuity account compounds interest monthly.

How much will Sarita have to deposit each month if she expects the computer to cost $2,800 at the end of two years.

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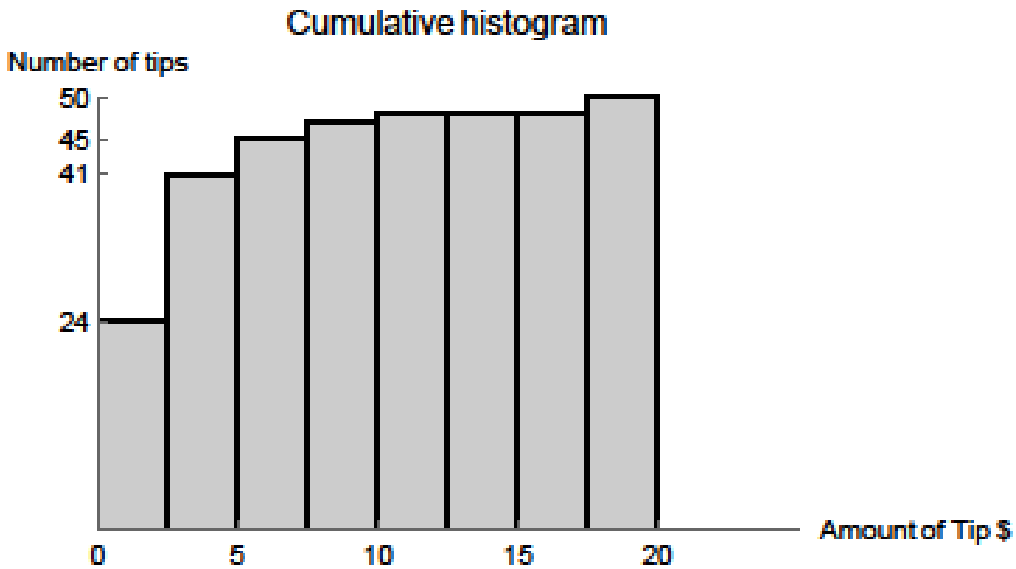
Question 31 (4 marks)

Peter works in a restaurant, and each customer can leave him a tip, some extra money, for good service. On one day, Peter earns 50 tips, as shown in the cumulative histogram:



One of the tips was incorrectly listed as $6, instead of $16.

Giving reasons, explain what effect would this have on each the following statistics?

(a) The median: **2**

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(b) The mean: **2**

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Question 32 (6 marks)

In order to set up a mining operation, a company sets up a trust account to make annual payments towards the construction and maintenance of infrastructure in the surrounding community. The account earns compound interest at 3% per annum.

Initially, the company deposits \$5 million into the account, and makes equal annual withdrawals of $\$R$, at the start of each year, beginning immediately.

(a) Show that at the end of two years, the amount left in the account is given by $A_2 = 5,000,000 \times 1.03^2 - R \times 1.03(1.03+1)$. **2**

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(b) Show that the amount of money left in the account at the end of n years is given by the expression $A_n = 5,000,000 \times 1.03^n - 103R\dfrac{(1.03^n - 1)}{3}$. **2**

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Question 32 continues on the next page

Question 32 (continued)

(c) If the company promises to provide the funds for 20 years, what is the maximum annual payment that can be made from the trust account? **2**

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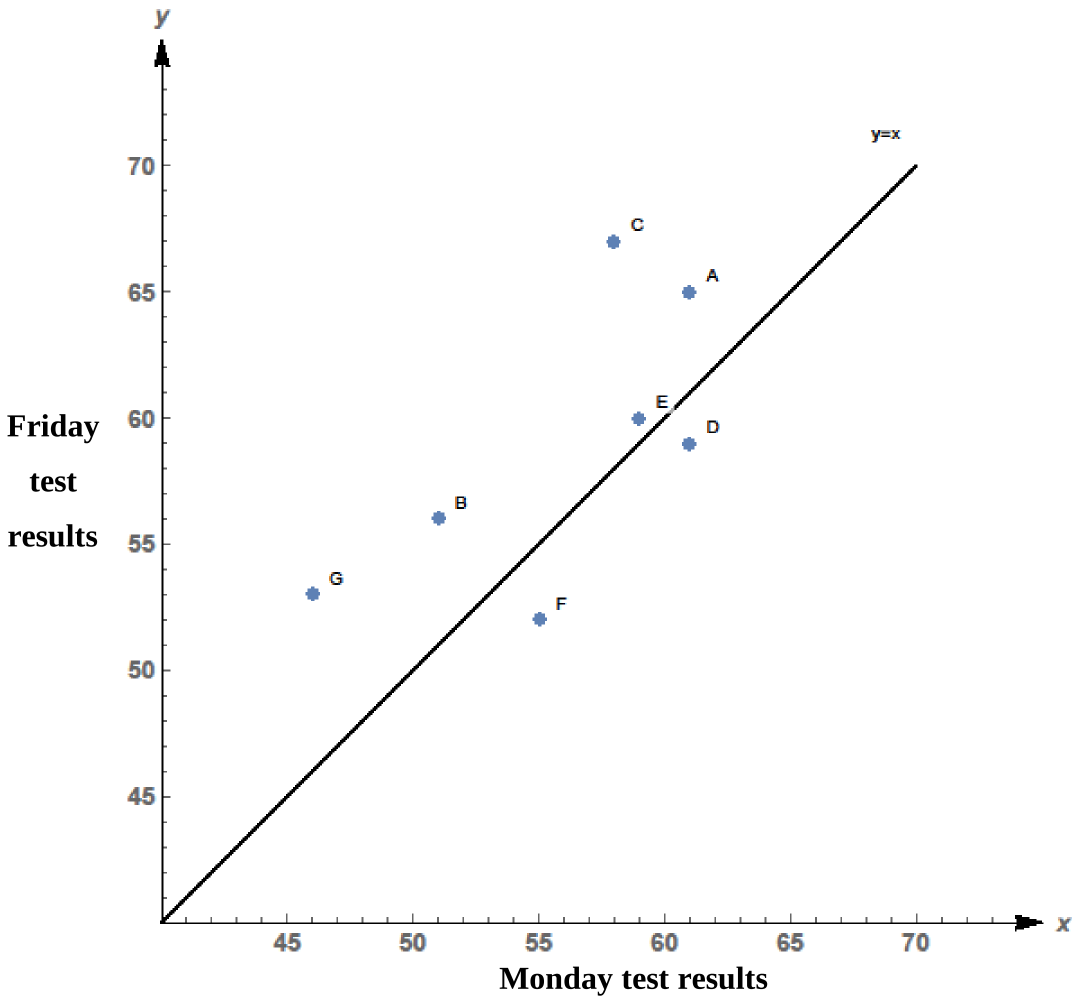
Question 33 (4 marks)

After a class of ten students sits a test on Monday, the teacher spends time doing revision and tests the students again on Friday. Only nine of the students are present for the second test on Friday. The results, in percentages, for those nine students, are shown in the following table.

Student	A	B	C	D	E	F	G	H	I	J
Monday [M]	61	51	58	61	59	55	46	58	62	63
Friday [F]	65	56	67	59	60	52	53	61	60	Absent

(a) Add the results of students H and I to the scatter plot. **1**



Question 33 continues on the next page

Question 33 (Continued)

(b) Determine the equation of the least squares-regression line of best fit. **1**

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(c) Using your answer to part (b), predict the result for the student who was absent for Friday's test. Give your answer to the nearest whole mark. **1**

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(d) The scatterplot above also shows the line $y = x$. Explain the significance of a student being represented below the line $y = x$. **1**

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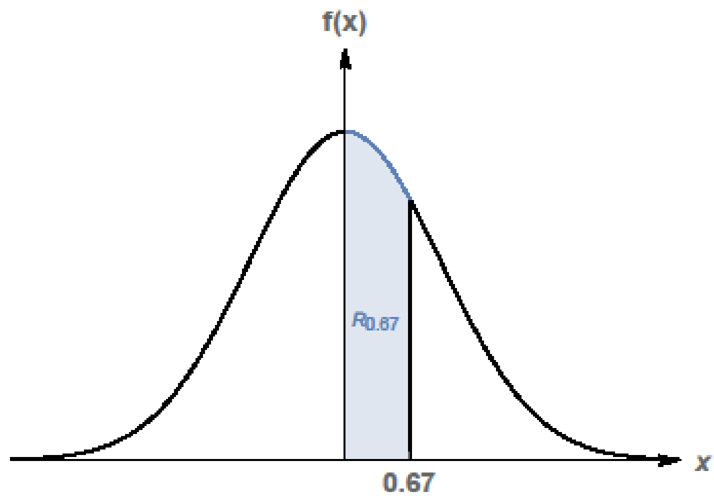
Question 34 (3 marks)

The random variable X has the standard normal distribution, with probability density function $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ over the domain $(-\infty, \infty)$. **3**

The graph of $y = f(x)$ is shown, and indicates a region $R_{0.67}$ which indicates the area under $y = f(x)$ over the domain $[0, 0.67]$.



If the area of $R_{0.67}$ is 0.25, determine the range of scores that could be identified as outliers.

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Section II extra writing space

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.